

Lie Theory: Chapter 1 Homework

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1. Let $\mathbb{R}_\vee^3$ be the Lie algebra defined in Section 1.2. Prove that if $x \cdot y$ denotes the dot product of vectors $x, y \in \mathbb{R}^3$, then

$$x \wedge (y \wedge z) = (x \cdot z)y - (x \cdot y)z \quad \forall x, y \in \mathbb{R}^3.$$

Proof. Observe that

$$\begin{aligned} x \wedge (y \wedge z) &= (x_1, x_2, x_3) \wedge (y_2 z_3 - y_3 z_2, y_3 z_1 - y_1 z_3, y_1 z_2 - y_2 z_1) \\ &= (x_2 y_1 z_2 - x_2 y_2 z_1 - x_3 y_3 z_1 + x_3 y_1 z_3, \\ &\quad x_3 y_2 z_3 - x_3 y_3 z_2 - x_1 y_1 z_2 + x_1 y_2 z_1, \\ &\quad x_1 y_3 z_1 - x_1 y_1 z_3 - x_2 y_2 z_3 + x_2 y_3 z_2) \\ &= (x_2 y_1 z_2 - x_2 y_2 z_1 - x_3 y_3 z_1 + x_3 y_1 z_3, \\ &\quad x_3 y_2 z_3 - x_3 y_3 z_2 - x_1 y_1 z_2 + x_1 y_2 z_1, \\ &\quad x_1 y_3 z_1 - x_1 y_1 z_3 - x_2 y_2 z_3 + x_2 y_3 z_2) \end{aligned}$$

and

$$\begin{aligned} &= (x \cdot z)y - (x \cdot y)z \\ &= (x_1 z_1 + x_2 z_2 + x_3 z_3)(y_1, y_2, y_3) - (x_1 y_1 + x_2 y_2 + x_3 y_3)(z_1, z_2, z_3) \\ &= (x_1 z_1 y_1 + x_2 z_2 y_1 + x_3 z_3 y_1, x_1 z_1 y_2 + x_2 z_2 y_2 + x_3 z_3 y_2, x_1 z_1 y_3 + x_2 z_2 y_3 + x_3 z_3 y_3) \\ &\quad - (x_1 y_1 z_1 + x_2 y_2 z_1 + x_3 y_3 z_1, x_1 y_1 z_2 + x_2 y_2 z_2 + x_3 y_3 z_2, x_1 y_1 z_3 + x_2 y_2 z_3 + x_3 y_3 z_3) \\ &= (x_2 y_1 z_2 - x_2 y_2 z_1 - x_3 y_3 z_1 + x_3 y_1 z_3, \\ &\quad x_3 y_2 z_3 - x_3 y_3 z_2 - x_1 y_1 z_2 + x_1 y_2 z_1, \\ &\quad x_1 y_3 z_1 - x_1 y_1 z_3 - x_2 y_2 z_3 + x_2 y_3 z_2), \end{aligned}$$

where the the last line comes after commuting elements in the terms with same index across all three variables (i.e. $x_1 z_1 y_1$ becomes $x_1 y_1 z_1$) since $\mathbb{R}$ is commutative and then cancelling them out.

We can see that both expressions reduce to the same final expression, so they are equal. $\square$

2. Check that

$$[e_{ij}, e_{kl}] = \delta_{jk} e_{il} - \delta_{il} e_{kj},$$

where δ is the Kronecker delta, defined by $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ otherwise.

Proof. Recall that $[e_{ij}, e_{kl}] = e_{ij}e_{kl} - e_{kl}e_{ij}$. The first term will have zeroes in all but the i^{th} row and zeroes in all but the l^{th} column since the only nonzero row in e_{ij} is the i^{th} one and the only nonzero column in e_{kl} is the l^{th} one. Thus the only possibly-nonzero entry is the one in the i^{th} row and l^{th} column. This will be 1 if and only if $j = k$ (and 0 otherwise) since we will then have the 1 from e_{ij} multiplied by the 1 in e_{kl} and zeroes everywhere else when doing the matrix multiplication. Thus we obtain the first term $\delta_{jk}e_{il}$ of our desired result.

The only other term to deal with is the second one, for which we use the same argumentation on $-e_{kl}e_{ij}$ to conclude that we subtract a 1 from the entry at the j^{th} column of the k^{th} row if and only if $i = l$, and a zero otherwise. This gives us our second term of $-\delta_{il}e_{kj}$, and we are done. $\square$

3. Prove that the set e_{ij} for $i \neq j$ together with the set $e_{ii} - e_{i+1, i+1}$ for $1 \leq i < n$ forms a basis for $\mathfrak{sl}(n, F)$.

4. Prove that an ideal of a Lie algebra is always a subalgebra.

Proof. Let I be an ideal of a Lie algebra L . Then I is a linear subspace of L by virtue of being an ideal, so we just need to show that $[x, y] \in I$ whenever $x, y \in I$. This is quite trivial, since I being an ideal guarantees that for any $x \in L$ and $y \in I$, $[x, y] \in I$, and surely $x \in L$ if $x \in I \subset L$, so there is nothing more to show. $\square$

5. Consider the adjoint homomorphism map defined (on page 4) for all $x, y \in L$ by

$$(\text{ad } x)(y) := [x, y].$$

- (a) Prove that ad is linear.
- (b) Prove that ad is a homomorphism.
- (c) Prove that the kernel of ad is the center of L .

Proof.

- (a) For $a, b \in F$ and $x, y, z \in L$,

$$\begin{aligned} (\text{ad}(ax + bz))(y) &= [ax + bz, y] \\ &= a[x, y] + b[z, y] \quad (\text{by bilinearity of the bracket}) \\ &= a(\text{ad } x)(y) + b(\text{ad } z)(y). \end{aligned}$$

- (b) We have already shown that ad is linear, so we just need to show that $\text{ad}[x, y] = [\text{ad } x, \text{ad } y]$. Indeed, for any $x, y, z \in L$

$$\begin{aligned} (\text{ad}[x, y])(z) &= [[x, y], z] \\ &= -[z, [x, y]] \\ &= [x, [y, z]] + [y, [z, x]] \\ &= [x, [y, z]] - [y, [x, z]] \\ &= (\text{ad } x)([y, z]) - (\text{ad } y)([x, z]) \\ &= (\text{ad } x)((\text{ad } y)(z)) - (\text{ad } y)((\text{ad } x)(z)) \\ &= ((\text{ad } x) \circ (\text{ad } y))(z) - ((\text{ad } y) \circ (\text{ad } x))(z) \\ &= [\text{ad } x, \text{ad } y](z). \end{aligned}$$

- (c) The center of L is defined as the set of elements x of L that satisfy $[x, y] = 0$ for any $y \in L$. If $x \in L$ is in the kernel of ad , then $\text{ad } x$ is the zero map, so by definition $0 = (\text{ad } x)(y) = [x, y]$ for all $y \in L$. Thus elements of the kernel of ad are in the center, and just as easily we see that if x is in the center of L , $[x, y] = 0$ for all $y \in L$, so $\text{ad } x$ is the zero map. By mutual inclusion then, the kernel of ad is the center of L .

□

6. Prove that $\text{Der}(A)$ is a vector subspace of $\text{gl}(A)$.

Proof. By definition, derivations are linear, so $\text{Der}(A) \subset \text{gl}(A)$. We need to show, then, that given an algebra A over a field F along with derivations $D, E \in \text{Der}(A)$ and $c \in F$, $D + E \in \text{Der}(A)$ and $cD \in \text{Der}(A)$. Indeed, for any $a, b \in A$, we have

$$\begin{aligned} (D + E)(ab) &= D(ab) + E(ab) \\ &= aD(b) + D(a)b + aE(b) + E(a)b \\ &= aD(b) + aE(b) + D(a)b + E(a)b \\ &= a(D(b) + E(b)) + (D(a) + E(a))b \\ &= a(D + E)(b) + (D + E)(a)b \end{aligned}$$

and

$$\begin{aligned} (cD)(ab) &= c(D(ab)) \\ &= c(aD(b) + D(a)b) \\ &= c(aD(b)) + c(D(a)b) \\ &= a(cD(b)) + (cD(a))b \quad (\text{by bilinearity of algebra multiplication}) \\ &= a(cD)(b) + (cD)(a)b, \end{aligned}$$

so both satisfy the Leibniz rule and are thus derivations as desired.

□

7. Let L_1 and L_2 be two abelian Lie algebras. Show that L_1 and L_2 are isomorphic if and only if they have the same dimension.

Proof. Suppose L_1 and L_2 both have dimension n , and let $\{x_1, \dots, x_n\}$ and $\{y_1, \dots, y_n\}$ be bases of L_1 and L_2 respectively. Then we can define a function $\varphi : L_1 \rightarrow L_2$ by letting $\varphi(x_i) = y_i$ for each i , and

$$\begin{aligned} \varphi(a_1x_1 + \dots + a_nx_n) &:= a_1\varphi(x_1) + \dots + a_n\varphi(x_n) \\ &= a_1y_1 + \dots + a_ny_n. \end{aligned}$$

This is well-defined since every element of L_1 can be expressed uniquely as a linear combination of $\{x_1, \dots, x_n\}$. It is also linear by construction, it is injective since linear combinations of $\{y_1, \dots, y_n\}$ are unique, and it is surjective because *every* element of L_2 can be expressed as such a linear combination. That leaves us to confirm that $\varphi([x, y]) = [\varphi(x), \varphi(y)]$. Indeed,

given $x, y \in L_1$,

$$\begin{aligned}
& [\varphi(a_1x_1 + \cdots + a_nx_n), \varphi(b_1x_1 + \cdots + b_nx_n)] \\
&= [a_1y_1 + \cdots + a_ny_n, b_1y_1 + \cdots + b_ny_n] \\
&= a_1[y_1, b_1y_1 + \cdots + b_ny_n] + \cdots + a_n[y_n, b_1y_1 + \cdots + b_ny_n] \\
&= (a_1b_1[y_1, y_1] + a_1b_2[y_1, y_2] + \cdots + a_1b_n[y_1, y_n]) + \cdots + (a_nb_1[y_n, y_1] + a_nb_2[y_n, y_2] + \cdots + a_nb_n[y_n, y_n]) \\
&= \cdots \\
&= \varphi([a_1x_1 + \cdots + a_nx_n, b_1x_1 + \cdots + b_nx_n]) \\
&= \varphi([a_1x_1 + \cdots + a_nx_n, b_1x_1 + \cdots + b_nx_n])
\end{aligned}$$

□

8. Find the structure constants of $\mathfrak{sl}(2, F)$ with respect to the basis given by the matrices

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

9. Prove that $\mathfrak{sl}(2, \mathbb{C})$ has no non-trivial ideals.

10. Let S be an $n \times n$ matrix with entries in a field F . Define

$$\mathfrak{gl}_S(n, F) = \{x \in \mathfrak{gl}(n, F) : x^T S = -Sx\}.$$

(a) Show that $\mathfrak{gl}_S(n, F)$ is a Lie subalgebra of $\mathfrak{gl}(n, F)$.

(b) Find $\mathfrak{gl}_S(2, \mathbb{R})$ if $S = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$.

11. Let A be an algebra and let $\delta : A \rightarrow A$ be a derivation. Prove that δ satisfies the Leibniz Rule

$$\delta^n(xy) = \sum_{r=0}^n \binom{n}{r} \delta^r(x) \delta^{n-r}(y) \quad \forall x, y \in A.$$